

AE-362: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 5 mM KOH

Date: 2024-10-17

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

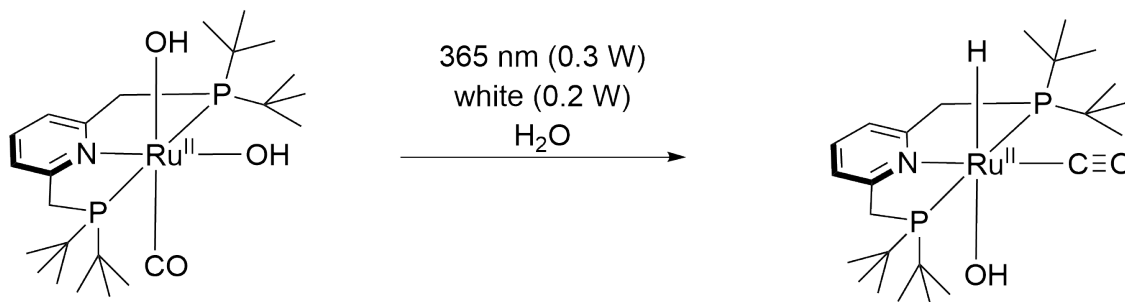
Category: Two-photon

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	AE-348: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, water, longer irradiation Two-photon - AE-357: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h (AE-310-1)	4.48	/	2.50	2.40	558.65	/
KOH (5 mM)	Prep work - AE-361: 5 mM Stocksolution of KOH in water	/	/	/	/	/	0.9

Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
17.10		A 10 mL flask was prepared	
	13:30	To the flask [Ru] was added under air	
		The flask was set under argon	
	14:45	To the flask KOH (5 mM) was added	
	14:55	The obtained mixture was stirred at 320 rpm.	left.jpg
	15:05	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-362-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	AE-362-1.jpg
21.10	10:20	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup.jpg
	10:32	The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	10:33 - 13:33	The J.Young NMR tube was irradiated for 3:00 h	irrad setup 1.jpg irrad setup 2.jpg
	16:50	An NMR was measured (AE-362-3)	AE-362-3.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
18.10	10:56	AE-362-1	³¹ P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-362-1.zip	AE-362-3_101.nmrium	/	as expected
22.10	18:56	AE-362-3	³¹ P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-362-3.zip	AE-362-3_101.nmrium	Two-photon - AE-349: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH, shorter irradi (AE-349-3)	prod formed, but significant amounts of substrate left

Results

Sample	Yield (hydride species) [%]	Yield (dihydroxo) [%]	Data
AE-362-3	22	23	AE-362-3_Yield_Determination.xlsx

For comparison with other experiments see: [Two-photon - AE-371: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 0.1 mM KOH](#)

Linked experiments

- [AEI-022: Radiation of \[Ru\(CO\)\(OH\)2\(PNN\)\], NO2-route, 365 nm and white LED](#)
- [AEI-026: Capillary production](#)
- [AEI-064: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AEI-067: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], crude, 365 nm and white](#)
- [AEI-071: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purer, 365 nm and white](#)
- [AE-105: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AE-110: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white](#)
- [AE-119: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 52 mM KOH, 365 nm and white](#)
- [AE-140: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white, longer duration I](#)
- [AE-145: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AE-158: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white](#)

- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Prep work - AE-361: 5 mM Stocksolution of KOH in water

Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation

Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-371: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 0.1 mM KOH

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-362-3_Yield_Determination.xlsx

sha256: 182e059ec11229770ecc2a208a3ab92a50ec1f936f7f35e4cd0160c1fcbb80db

AE-362-3_101.nmrium

sha256: 052f1397eeef1a8083551b95912c028edc20b42afee85a1e9612a91fd76484b9

AE-362-3.zip

sha256: c5e13254201f0d065d1a6f951e19354f55dc8b00ee2d11f2b00eb3bcaccb571c

AE-362-1.zip

sha256: 07dc43e53a0a7a2848198a2d0b68fc99364ae6bd09c027447590a7cea6e6b13e

irrad-setup-2.jpg

sha256: 747d6767dc314835645062a3add2b6935202e61177f52fced05e671803877bc1



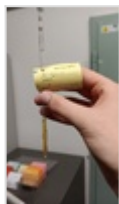
irrad-setup-1.jpg

sha256: a2a6dfaf1d38731f9aecfb4f653f1379321510d6e68d93ef46d38a506ea4e54d



AE-362-3.jpg

sha256: aa8016f2d7204f6863745fea3ed747b0b143802624116f1570a23d02c51ccff7



power-setup.jpg

sha256: de32d5c73382019c9f2e0c21ecdbe3f8c1c3813392cedc75482be854db6aa265



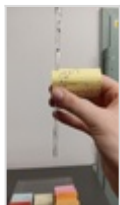
left.jpg

sha256: 1f767bce0a24d83e3f9c21376dc18e1fec3a587507c2285f3e023449a33058e4



AE-362-1.jpg

sha256: 80d01b43136d0c02a09d70a86cb7eae90ddf25604fcffa5ec9e1e5e20120b5d1



AEI-06711.cdxml

sha256: 65d02cd9f4a21042d9bbb3b568aacc15b4f98db9ed95de74a4e781b0391592



Unique eLabID: 20241018-a5bebd3410e035453e806290bd832b31da4c4efc
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1431>